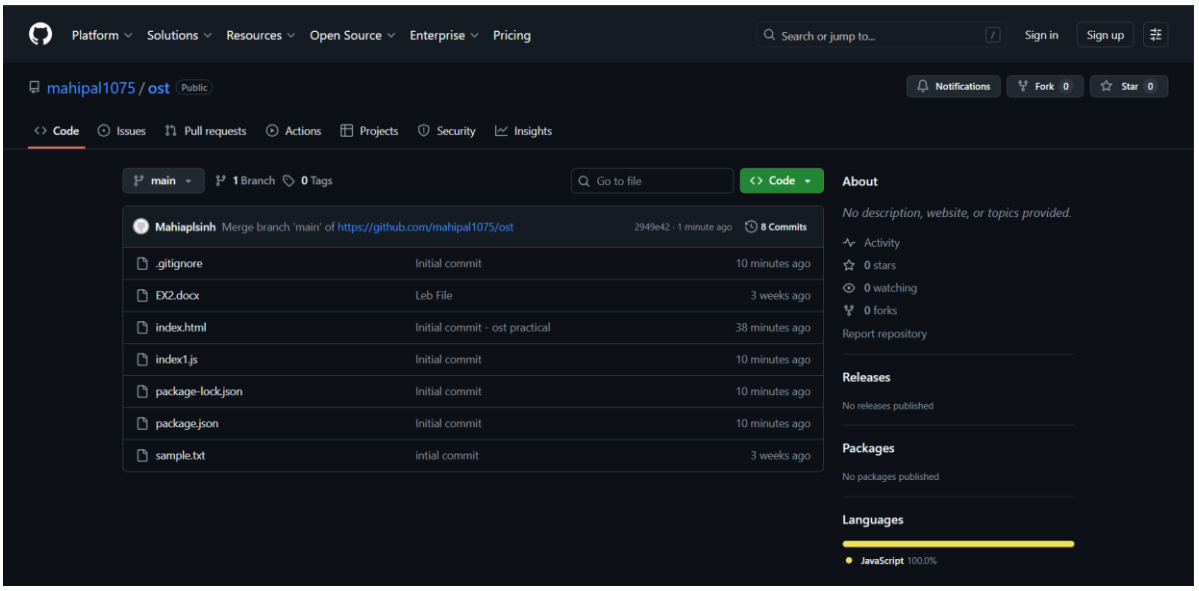
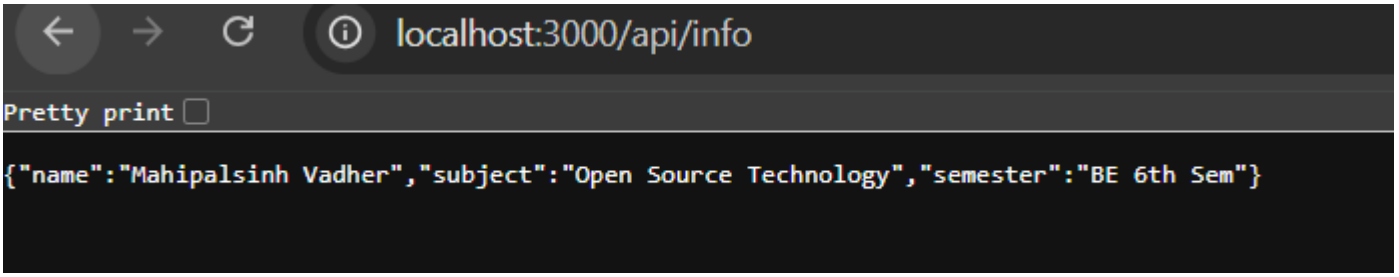
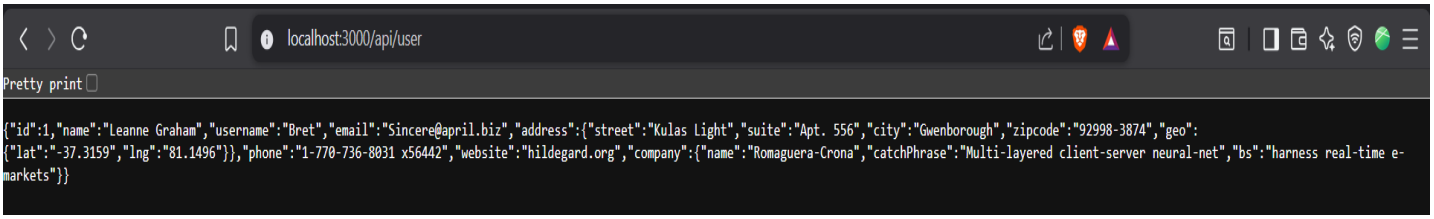
Prectical 4

Aim: Use npm / yarn / pip to install and manage packages

```
Step 1: Check Installation

C:\Users\Mahipal>node -v
v22.14.0

C:\Users\Mahipal>npm -v
11.7.0

Step 2: Create New npm Project
D:\> mkdir npm-lab
D:\> cd npm-lab
D:\npm-lab>npm init -y

Step 3: Install Required Packages
D:\npm-lab>npm install express

added 65 packages, and audited 66 packages in 4s

22 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\npm-lab>npm install axios

added 11 packages, and audited 77 packages in 4s

24 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\npm-lab>npm install nodemon --save-dev

added 27 packages, and audited 104 packages in 4s

28 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

D:\npm-lab>npm list
npm-lab@1.0.0 D:\npm-lab
├── axios@1.13.5
├── express@5.2.1
└── nodemon@3.1.11

Step 4: Create Web Server
const express = require("express");
const axios = require("axios");
const app = express();
```

```
app.get("/", (req, res) => {
res.send("Hello Open Source Lab");
});

// JSON Route
app.get("/api/info", (req, res) => {
res.json({
name: "Mahipalsinh Vadher",
subject: "Open Source Technology",
semester: "BE 6th Sem"
});
});

// Axios Demo Route
app.get("/api/user", async (req, res) => {
const response = await axios.get("https://jsonplaceholder.typicode.com/users/1");
res.json(response.data);
});

app.listen(3000, () => {
console.log("Server running at http://localhost:3000");
});

Step 5: Run Server (Normal Way)

D:\npm-lab>node index.js
Server running at http://localhost:3000

Step 6: Add npm Scripts
Open package.json and modify scripts section:

"scripts": {
"start": "node index.js",
"dev": "nodemon index.js"
}

Step 7: Run Using npm Script

D:\npm-lab>npm run dev

> npm-lab@1.0.0 dev
> nodemon index.js

[nodemon] 3.1.11
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node index.js`
Server running at http://localhost:3000
Terminate batch job (Y/N)? y

D:\npm-lab>npm run dev

> npm-lab@1.0.0 dev
> nodemon index.js
```

```
[nodemon] 3.1.11
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node index.js`
Server running at http://localhost:3000

Step 8: Delete node_modules and Reinstall

Delete manually
Now reinstall:
D:\npm-lab>npm install

added 103 packages, and audited 104 packages in 5s

28 packages are looking for funding
  run `npm fund` for details

Step 9: Add .gitignore File

Create file: .gitignore
Add:
node_modules/

Step 10: Upload Project to Git
D:\npm-lab>git init
Reinitialized existing Git repository in D:/npm-lab/.git/

D:\npm-lab>git add .
warning: in the working copy of 'index1.js', LF will be replaced by CRLF the next time
Git touches it

D:\npm-lab>git commit -m "Initial commit"
[master (root-commit) 8534dfc] Initial commit
 4 files changed, 1358 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 index1.js
 create mode 100644 package-lock.json
 create mode 100644 package.json

D:\npm-lab>git remote add origin https://github.com/mahipal1075/ost

D:\npm-lab>git branch -M main D:\npm-lab>git push origin main --force
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 12.29 KiB | 2.46 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mahipal1075/ost
 + 5cc42a1...8534dfc main -> main (forced update)
```



Hello Open Source Lab